

A partial order toolbox for building synthetic indicators of sustainability with ordinal data

Marco Fattore^{a,*}, Leonardo Salvatore Alaimo^{b,1}

^a University of Milano-Bicocca, Italy

^b University of Rome-La Sapienza, Italy

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ABSTRACT

This paper designs and implements a statistical toolbox for the construction of synthetic indicators of sustainability and rankings of multidimensional systems of ordinal variables. The toolbox employs results from partial order theory to provide a purely ordinal way to perform the classical steps of indicator construction (i.e., with neither quantification nor aggregation of the input variables) and thus fills a gap in the statistical literature. “Ordinal” non-aggregative procedures are developed to score statistical units and assess the reliability of the final rankings as well as account for exogenous information relating to variable importance. The toolbox is introduced and shown in action through the real example of Sustainable Development Goal 16 (Peace, Justice, and Strong Institutions), specifically as it relates to Political Pluralism and Participation in North and South American countries.

1. Introduction

The main goal of this paper is to develop a statistical toolbox for the construction of synthetic indicators and the extraction of rankings from ordinal multi-indicator systems (MISes) to support a multi-criteria assessment of sustainability and the degree to which the Sustainable Development Goals (SDGs) have been achieved. This topic is relevant from both a methodological and an applied point of view, given the lack of well-founded statistical procedures that are able to handle ordinal MISes and the growing demand for synthetic information by scholars, decision-makers, and institutions in different disciplines and contexts (the proposed methodology is also suitable for cardinal data [1–5] and even for indicators scored against different scale types [6]). For example, in the assessment of nations’ wealth, much attention is devoted to people’s well-being and quality of life, which are often measured by gathering qualitative, ordinal, and multidimensional information on individuals’ socioeconomic status and personal life. Psychometric scales are ordinal in nature, like those employed in many evaluation processes from, for example, multidimensional deprivation to schooling attainments or customer satisfaction. However, while statistics offers a great variety of tools for the effective treatment of cardinal data systems, the ordinal case is still devoid of a sound and comprehensive statistical toolbox. As is extensively discussed in the following section, classical statistics often addresses ordinal data as rough manifestations

of underlying unobserved cardinal variables to be recovered and reconstructed, thus leading the construction of synthetic indicators back to the tools usually used in multidimensional data analysis. This point of view can be criticized in many respects [7,8], in particular because many socioeconomic traits, being nuanced and “fuzzy”, are *naturally* described ordinally rather than cardinally (see, e.g., [9]). In addition, it must be noticed that the tendency to replace ordinal data with numerical data is, historically, partly due to the supposed lack of a proper mathematical language that can be used to uniformly treat this type of input. In fact, however, such a language does exist and can be fruitfully employed to develop new statistical tools for ordinal data treatment, as we do in the present paper for indicators of sustainability. Here, the entire scoring process is done in the framework of *partial order theory* [10, 11], a branch of discrete mathematics designed to deal with order relations that provides an ordinal way to perform the classical steps of indicator construction. In brief, the idea is to synthesize the input data by multidimensionally comparing the statistical units and quantifying the extent to which they tend to dominate each other, rather than by aggregating the units’ scores on the input variable system into numbers to be used for ranking (which cannot be pursued on non-numerical data). In other words, we “quantify multidimensional comparisons”, rather than “comparing aggregated quantifications”, thereby exploiting the information contained in the order relation set on the data and fully respecting their non-numerical nature. We develop this approach

* Corresponding author.

E-mail address: marco.fattore@unimib.it (M. Fattore).

¹ Joint work and that both have contributed equally to it.

by using data pertaining to Sustainable Development Goal (SDG) 16 (*Peace, Justice, and Strong Institutions*), namely Political Pluralism and Participation in North and South American countries, as a leading example. We show that comparing units assessed against multi-indicator systems leads us to naturally organize the data into a component-wise *partially ordered set* (or *poset*, for short), which is assumed as the basic data structure for the scoring and ranking process. Importantly, posets have both a “vertical” dimension (*comparability/dominance* between elements) as well as a “horizontal” one (*incomparability*), the latter accounting for the structural differences between the units in the MIS. Both must be considered to obtain a faithful synthesis of the data. For each poset element x_j , we first define its so-called *dominance score*, which is essentially a fuzzy measure of the cardinality of the downset of x_j (i.e., of the set of elements “below” it), thus providing a natural criterion for ranking units. Dominance scores are complemented with measures of incomparability as a way to assess the degree of “rankability” of the units and to check whether putting them into a final ranking is reliable or forced. We then observe that, in addition to identifying the units in the population/sample, the input MIS provides also the proper “scoring context” by component-wise ordering both the observed and unobserved achievement patterns derived from the indicator system. Embedding the computations in this larger space makes them much more informative and leads to a definition of *separation scores* that, unlike the dominance ones, incorporate information on the “distance” between units in the poset, thereby greatly improving the discrimination power of the scoring process. In concrete applications, it is often the case that the input variables have different degrees of importance and the issue of how to account for this in an ordinal context naturally arises. In this paper, this is done by modifying the structure of the component-wise poset based on the priorities assigned to the variables and using the new poset as the input for the computation of the scores. Finally, we generalize the entire framework by attaching frequencies and weights to the poset elements.

This paper is not particularly challenging from a technical and mathematical point of view, but the methods and results are nontrivial. While the computation of the final synthetic scores is performed through simple sums (or, possibly, through Singular Value Decomposition (SVD)-based aggregations) of elementary scores, the conceptual and formal setting underlying this final step is quite articulated and subtle, as it exploits and integrates concepts and tools from order theory that are not mainstream in the statistical literature and that call for a new perspective on the indicator construction process. In any case, the toolbox can be quite easily implemented in high-level programming languages and provides a practical statistical platform for concrete applications.

The tools designed in the following sections not only fulfill methodological and technical gaps in the literature, but also provide a more reliable and effective way to convey information to policymakers. Composite indicators [12] are increasingly used for policy design and evaluation, yet they are deeply criticized for their aggregative-compensative nature, their tendency to mix “apples and oranges”, and for the lack of measurement models supporting them (see [13] and the discussion in the next section). As a result, the interpretation and reliability of the final indicators are often controversial, which makes their use in practice problematic [14,15]. When dealing with ordinal data, these concerns are even more worrying because of the way the input data must be treated to fit into the composite scheme (see, e.g., [16]). The posetric approach, owing to its non-aggregative nature, overcomes most of these issues by being more consistent with the input data and putting much weaker and less artificial hypotheses on them. Notwithstanding its main methodological nature, this paper also has applicative value and should be considered as supporting reference material for policy-making.

This paper produces new results and combines them with those of some existing studies, thus providing an original and comprehensive exposition of the posetric approach to scoring and ranking. The remaining text is organized as follows. Section 2 provides a brief

outline of the literature; Section 3 introduces the Political Pluralism and Participation example and the first partial order concepts; Section 4 develops the computation of the dominance and incomparability scores for a generic input poset; Section 5 specializes it for posets deriving from multi-indicator systems and introduces separation scores; Section 6 demonstrates how to account for variable importance; Section 7 generalizes the computation of the scores when weights are attached to the poset elements; Section 8 provides a general discussion of the main results developed in the paper; Section 9 concludes and outlines prospects for future research. A short mathematical Appendix provides the proof of the optimality of the “variable importance map” introduced in Section 6.

2. Literature review

The statistical literature on indicator construction and ranking extraction is extremely vast and thus only an outline is provided here. Most of this literature deals with the synthesis of quantitative data and the computation of composite indicators [17,18] — that is, of weighted variable aggregations to “condense”, in a formative way, the input indicator system into a unidimensional score vector and possibly extract a final ranking. Composites are among the most commonly employed tools for evaluation and decision-making in many areas of applied statistics (and in sustainability assessment as well; see, e.g., [19–23]), although they have often been criticized for their compensative nature (both epistemologically [13] and as practical tools), the absence of neat criteria for the choice of weighting system (see, e.g., [24]), and their tendency to mix “apples and oranges”. This debate has been ongoing in the literature for many years (see, e.g., [25–28]). In this respect, reflective models such as factor analysis and its structural equation generalizations [29] are often advocated as a firmer alternative. Although neater in their foundations, they suffer from the well-known indeterminacy problem [29,30], which makes them effective for the structural analysis of correlation structures, but less so for the aggregative quantification of unobservable variables. Dimensionality reduction tools are also often employed in indicator construction to exploit the covariance structure of the input data without using underlying inferential models, as in the case of the well-known Singular Value Decomposition/Principal Component algorithm.

Despite their deep differences, all of these approaches share the “synthesis as variable aggregation” perspective and ultimately build the final indicator as a linear combination of the input variables. When dealing with ordinal data, this strategy cannot be directly pursued and alternative algorithms must be designed to treat non-numerical objects. The mainstream literature focuses on scaling and quantification procedures to turn the ordinal inputs into cardinal scores to be aggregated in the same spirit as in the quantitative case. This approach ranges from simply coding ordinal variables as numbers (see, e.g., [31] p. 311) to sophisticated algorithms that compute numerical scores based on optimization criteria, as in Categorical and Nonlinear Principal Component Analysis, Multiple Correspondence Analysis, Homogeneity Analysis, and various similar tools (see [32] and, in particular, [33] for a comprehensive introduction to the so-called Gifi system). Most of the scaling procedures, however, are designed for general categorical variables and do not explicitly exploit the ordinal features of the data. There have been some attempts to build specific quantification algorithms in the ordinal case [34–36], but this topic is considerably less explored and developed. At times, and especially in the social sciences, ordinal variables are simply turned into binary attributes by introducing exogenous cut-offs, and final scores are obtained by counting procedures, as in the Alkire-Foster approach to the evaluation of multidimensional poverty [37].

An alternative stream of literature represents ordinal multi-indicator systems as partially ordered sets and builds synthetic indicators and rankings with no variable aggregation, by exploiting the information hidden in the structure of the partial order relation put on the data,

with the aim of more faithfully respecting their ordinal nature. The simplest way to do this is by computing so-called *average ranks* — that is, to consider the set $\Omega(\pi)$ of all of the linear orders compatible with the input poset π (see Section 4) and score each element based on its average position in them [6,38,39]. As an alternative, Patil and Taille [40] extract the final ranking based on the stochastic comparison of the distributions of the position of each element over the linear orders in $\Omega(\pi)$. Yet another approach [41] defines a distance between partially ordered sets and addresses the construction of rankings as a problem of unidimensional approximation to the input ordered structure in the spirit of dimensionality reduction. The idea of exploiting the partially ordered structure of the data rather than the set of variable scores is being increasingly employed, for example in environmental sciences [42] and in the construction of social indicators (see, e.g., [43–46]), where posets proved useful in accounting for the nuances and multifaceted nature of socioeconomic traits. Interestingly, these procedures are not limited to posets deriving from multi-indicator systems, but can be applied to any partially ordered structure and have converged into a different research area pertaining to scoring and ranking preference data, as is typical in social choice analysis, multi-criteria decision-making and tournament theory (see, e.g., the works of Saaty [47–49], Todeschini [50] and Keener [51]). In fact, posetic procedures are scattered across the literature, but no systematic theory of their use in indicator construction is available and much work still needs to be done to develop a sound, comprehensive and effective algorithmic toolbox for applied data analysis and for sustainability evaluation and monitoring in particular. Taking a first step in this direction is one of the main aims of this paper.

3. Leading example: Political Pluralism and Participation in North and South American countries

We introduce and develop the scoring/ranking toolbox by considering data extracted from the *Freedom in the World 2022* dataset (for details, see <https://freedomhouse.org>) on the rights and freedoms enjoyed by individuals across the world. The dataset assesses 195 countries and 15 territories along three dimensions relating to political rights, namely, (A) *Electoral Processes*, (B) *Political Pluralism and Participation* and (C) *Functioning of Government*, and on four pillars relative to civil liberties, namely, (D) *Freedom of Expression and Belief*, (E) *Associational and Organizational Rights*, (F) *Rule of Law* and (G) *Personal Autonomy and Individual Rights*. For the sake of simplicity and to allow for easier graphical representations (particularly useful when dealing with ordered structures), here we exploit only the Political Pluralism and Participation pillar for 35 North and South American countries. This pillar is formed by four subpillars relating to (B1) *Right of people to organize in different political parties*, (B2) *Opportunity for the opposition to increase its support or gain power through elections*, (B3) *Freedom of people's political choices from domination by external forces* and (B4) *Full political rights and electoral opportunities for relevant social groups and population segments*; each subpillar is assessed on a 0–4 ordinal scale, which produces the restricted multi-indicator system (MIS) given in Table 1, where the Political Pluralism and Participation (PP&P) score configurations (*profiles*) of the selected countries are reported. As can be seen from the table, only 18 unique profiles are observed in the MIS, with some countries sharing the same pattern. Our final aim is to score and rank the countries according to their profiles. Since variables B1–B4 are ordinal and cannot be meaningfully summed or aggregated in any way, the only legitimate operation one can perform on the dataset is to compare PP&P profiles; since at this stage no variable is deemed more important than the others, this is naturally done *component-wise* according to the following general definition:

Definition. (Component-Wise Ordering). Let $x_1 = x_{11}x_{12} \dots x_{1k}$ and $x_2 = x_{21}x_{22} \dots x_{2k}$ be two k -dimensional ordinal profiles. We posit that x_2 component-wise dominates x_1 , written as $x_1 \triangleleft_x x_2$, if and only if

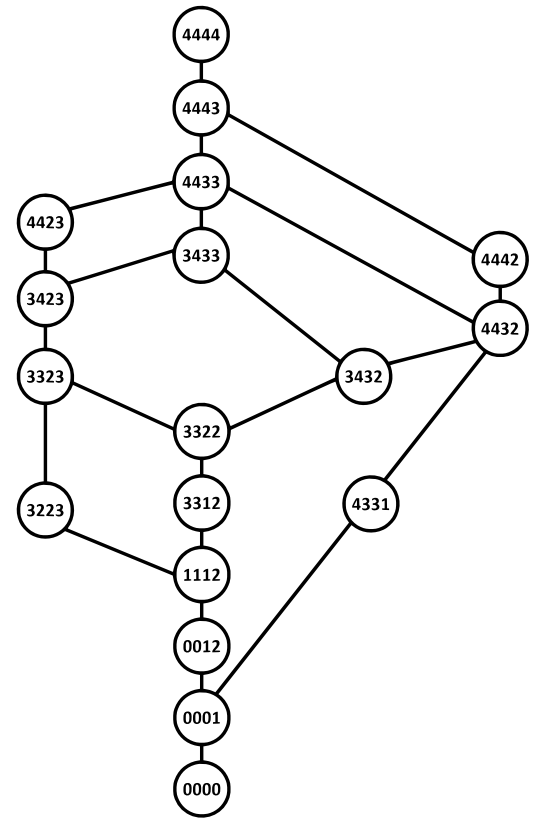


Fig. 1. Hasse diagram of π_x , the component-wise poset of the unique profiles derived from the PP&P MIS. The diagram must be read from top to bottom: if node y is linked by a downward path of edges to node x , then the two nodes are comparable and $x \triangleleft_x y$; if such a path does not exist, the two nodes are incomparable.

$x_{1i} \leq x_{2i}$ for $i = 1, \dots, k$ and j exists such that $x_{1j} < x_{2j}$, i.e., if and only if all the scores of x_2 are not lower than those of x_1 and at least one of them is higher (symbol “ $\triangleleft$ ” refers to the component-wise order being the so-called product order [10] of the input variables, seen as complete orders). We denote $x_1 \triangleleft_x x_2$ to include the case where $x_{1i} = x_{2i}$ for all $i = 1, \dots, k$.

If $x_1 \triangleleft_x x_2$ or $x_2 \triangleleft_x x_1$, the two profiles are said to be *comparable*, otherwise they are said to be *incomparable* (written as $x_1 \parallel x_2$).

Due to the presence of *conflicting scores*, some pairs of input profiles do not satisfy the above ordering criterion and are incomparable (consider, for example, the profiles of Belize and Bolivia), so the set P of unique profiles derived from the MIS (see Table 2) can be ordered only *partially*. The binary relation $\triangleleft_x$, which is *reflexive*, *antisymmetric* and *transitive* on P , is in fact called *partial order* [10] and the pair $\pi_x = (P, \triangleleft_x)$ is referred to as a *partially ordered set*, or a *poset*. Fig. 1 provides the Hasse diagram of π_x built on the 18 unique PP&P profiles. As can be seen, the PP&P poset has profile 4444 as its *maximum* or *top* (i.e., the element dominating all of the others) and profile 0000 as its *minimum* or *bottom* (i.e., the element dominated by all of the others) and comprises 125 comparabilities and 28 incomparabilities. Its longest *chain* (i.e., subset or, visually, downward sequence) of mutually comparable elements comprises 12 profiles, while the largest *antichains* (i.e., subsets of mutually incomparable elements) comprise 3 objects. The PP&P poset is thus said to have *height* equal to 12 and *width* equal to 3.

Our first goal is to design a statistical procedure that takes the poset π_x as input and delivers, for each of its elements, a synthetic PP&P *dominance* score and then a ranking of the countries. Given their ordinal nature, such a score cannot be obtained by the aggregation of the

Table 1

PP&P data for 35 American countries, assessed on a 0–4 ordinal scale (the higher the scores, the better the country achievements on the variables).

Country	B1	B2	B3	B4	Country	B1	B2	B3	B4
Ant. and Barb.	3	4	3	3	Guyana	4	4	3	2
Argentina	4	4	4	4	Haiti	1	1	1	2
Bahamas	4	4	4	4	Honduras	3	3	1	2
Barbados	4	4	4	4	Jamaica	4	4	2	3
Belize	4	4	4	2	Mexico	4	4	2	3
Bolivia	3	2	2	3	Nicaragua	0	0	1	2
Brazil	4	4	3	3	Panama	4	4	4	3
Canada	4	4	4	4	Paraguay	3	4	3	2
Chile	4	4	4	3	Peru	4	4	3	2
Colombia	3	3	2	3	St. Kitts and N.	4	4	4	3
Costa Rica	4	4	4	3	St. Lucia	4	4	4	4
Cuba	0	0	0	0	St. Vinc. and the G.	4	4	4	3
Dominica	4	4	4	4	Suriname	4	4	3	3
Dominican Rep.	4	3	3	1	Trin. and Tob.	3	4	3	3
Ecuador	3	4	3	3	United States	4	4	2	3
El Salvador	3	4	2	3	Uruguay	4	4	4	4
Grenada	4	4	4	4	Venezuela	0	0	0	1
Guatemala	3	3	2	2	/	/	/	/	/

Table 2

Unique profiles extracted from the PP&P MIS and their frequencies.

Prog.	1	2	3	4	5	6	7	8	9
Profile	4444	4443	4442	4433	4432	4423	4331	3433	3432
Freq.	8	5	1	2	2	3	1	3	1

Prog.	10	11	12	13	14	15	16	17	18
Profile	3423	3323	3322	3312	3223	1112	0012	0001	0000
Freq.	1	1	1	1	1	1	1	1	1

variables as in the case of composite indicators, but must be computed based on the *structure* of the partial order relation set on the profiles (i.e., of the network of comparabilities and incomparabilities in which they are embedded). In other words, profiles are scored by “quantifying multidimensional comparisons”, rather than by aggregating their internal ordinal configurations into numbers. The next section develops this logic to make it more precise and effective.

4. Dominance and incomparability scores

Since partial orders are binary relations, the most straightforward way to assign a ranking score to a profile x_j is to compute its *pairwise dominance degrees* over the other profiles and to aggregate these measures into a synthetic one. We thus introduce the following definition, holding for any partial order relation $\preceq$.

Definition. Let $\pi = (X, \preceq)$ be a finite poset over a set X . A function $pdom(\cdot, \cdot) : \pi \times \pi \rightarrow [0, 1]$ is called a *pairwise dominance function* (written $pdom(x_i, x_j) = pdom_{ij}$) if it satisfies the following properties:

1. $pdom_{ij} = 1$ if $x_i \prec x_j$;
2. $pdom_{ij} = 0$ if $x_j \preceq x_i$;
3. $pdom_{ij} = 1 - pdom_{ji}$ ($i \neq j$);
4. $pdom_{ij} \leq pdom_{ih}$ if $x_j \prec x_h$.

Properties 1 and 2 require that the degrees of dominance of x_j over the elements strictly dominated by it equal 1 and over the elements it is weakly dominated by equal 0 (i.e., no element dominates itself). Property 3 is a “consistency” requirement; the *monotonicity* of Property 4 requires $pdom$ to be *order-preserving* over the second argument when the first is kept fixed, which is a natural constraint for dominance functions; together with Property 3, it also leads to $pdom_{ij} \leq pdom_{hj}$ if $x_h \prec x_i$ (i.e., $pdom$ is *anti order-preserving* in its first argument). Notice that Properties 1 and 3 imply the strict part of Property 2, which has nevertheless been stated explicitly for the sake of clarity. Denoted by

n the number of profiles in π , function $pdom$ is naturally represented as an $n \times n$ *dominance matrix* D , with entries $D_{ij} = pdom_{ij}$. The entries of the j th column of D are the pairwise dominance scores of x_j over the other elements; those of the i th row are the scores by which x_i is dominated by the other profiles.

An infinite number of functions meet the above definition and several alternatives are possible. The simplest defines $pdom_{ij} = 1$ if $x_i \prec x_j$ and $pdom_{ij} = 0$ otherwise. The binary nature of this function, however, is unsuitable for our purposes since the scores of incomparable elements are all confounded into 0 and thus lose a great deal of information. For example, profiles 4432 and 3223 are incomparable in the Hasse diagram of Fig. 1, but the first represents a quite better PP&P configuration than the second, as it is dominated by fewer profiles and placed “nearer” to the top of the diagram; intuitively, 4432 should dominate 3223 “to a certain degree”, which the above choice does not reveal.

A much more informative function is provided by the so-called *mutual ranking probability* (mrp) of x_j over x_i [52], defined as

$$mrp_{ij} = \frac{|\lambda \in \Omega(\pi) : x_i \prec_\lambda x_j|}{|\Omega(\pi)|}, \quad (1)$$

where $\Omega(\pi)$ is the set of *linear extensions* of π , i.e. the set of *linear* (or *complete*, or *total*) orders λ of the elements of π such that $x_i \prec_\lambda x_j$ in π implies $x_i \prec_\lambda x_j$ in λ .

If $x_i \prec x_j$ in the input poset, then the same holds in each linear extension and so $mrp_{ij} = 1$ and $mrp_{ji} = 0$. If instead $x_i \parallel x_j$ in π , then both mrp_{ij} and mrp_{ji} take value in $(0, 1)$. The advantage of using mutual ranking probabilities is their capability to finely quantify the dominance degrees between incomparable elements. Unfortunately, for typical real posets, the number of linear extensions is so high that it is impossible to compute ranking probabilities exactly [52]. Even approximate computations based on sampling strategies [53] are highly demanding and often cannot be pursued effectively, thus limiting the class of posets that can be handled this way.

To compromise between information power and computational feasibility, Bruggemann, Lerche and Sørensen propose a dominance measure (bls) [39] based on easily computable features of the partial order relation. Its expression is complicated, but easy to implement computationally. It reads:

1. For $x_i \prec x_j$, $bls_{ij} = 1$ and $bls_{ji} = 0$;
2. For $x_i \parallel x_j$, $bls_{ij} = a_{ij}/(a_{ij} + a_{ji})$ where

$$a_{ij} = \frac{|Up_{ij}| + 1}{|Down_{ij}| + 1} \quad (2)$$

with

$$Up_{ij} = \{x_h : x_i \prec x_h\} \cap \{x_h : x_j \prec x_h\}^c \quad (3)$$

and

$$Down_{ij} = \{x_h : x_h \prec x_i\} \cap \{x_h : x_h \prec x_j\}^c \quad (4)$$

The bls function effectively resolves the dominance degrees of incomparable pairs (see Table 3, where the dominance of profile 4432 over profile 3223, i.e. $D_{5,14}$, is 0.93) and will be employed for the remainder of this paper.

By aggregating the bls pairwise scores, each poset profile x_j is then assigned a synthetic dominance score dom_j , which must satisfy the following two properties:

1. $x_i \prec x_j \Rightarrow dom_i < dom_j$;
2. $dom_j = 0 \Leftrightarrow x_j = \perp$ (i.e., the dominance score is null if and only if x_j is the bottom of the input poset).

The first is a key consistency requirement since the dominance score must increase as the poset element moves “upwards”, the second anchors the dominance scale to a reference point, chosen in this case as

Table 3

Dominance matrix of the PP&P poset, computed with the *bis* function (rows and columns report the indexes of the profiles, as given in Table 2).

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	1	1	0	.86	0	.33	0	.40	0	.16	.09	0	0	.03	0	0	0	0
4	1	1	.14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	1	1	1	1	0	.73	0	.80	0	.40	.25	0	0	.07	0	0	0	0
6	1	1	.67	1	.27	0	.04	.67	.06	0	0	0	0	0	0	0	0	0
7	1	1	1	1	1	.96	0	.96	.71	.88	.82	.40	.30	.64	.18	.09	0	0
8	1	1	.60	1	.20	.33	.04	0	0	0	0	0	0	0	0	0	0	0
9	1	1	1	1	1	.94	.29	1	0	.82	.67	0	0	.20	0	0	0	0
10	1	1	.84	1	.60	1	.13	1	.18	0	0	0	0	0	0	0	0	0
11	1	1	.90	1	.75	1	.18	1	.33	1	0	0	0	0	0	0	0	0
12	1	1	1	1	1	1	.6	1	1	1	1	0	0	.67	0	0	0	0
13	1	1	1	1	1	1	.7	1	1	1	1	1	0	.83	0	0	0	0
14	1	1	.97	1	.93	1	.36	1	.80	1	1	.33	.17	0	0	0	0	0
15	1	1	1	1	1	1	.81	1	1	1	1	1	1	1	0	0	0	0
16	1	1	1	1	1	1	.91	1	1	1	1	1	1	1	1	0	0	0
17	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0
18	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0

zero. A very simple, yet effective, choice for the dominance function is $dom_j = \sum_{i=1}^n pdom_{ij} = \sum_{i=1}^n D_{ij} = [\mathbf{1}_n^T D]_j$ (where $\mathbf{1}_n$ is the n -dimensional vector composed of 1s), which is obtained by summing the entries in the column of the dominance matrix corresponding to the selected poset element. Interpreting, in a fuzzy spirit, $pdom_{ij}$ as the “strength” of the statement $x_i \prec x_j$, dom_j is the “effective” number of poset elements dominated by x_j . Dominance scores attain their maximum when $pdom_{ij} = 1$ for $i \neq j$ (i.e., when x_j is the top $\top$ of the poset). In this case, $dom_{\top} = n - 1$ and $dom_j^* = dom_j / (n - 1)$ is a normalized dominance score that takes value in $[0, 1]$.

Turning the input poset π into a linear order may indeed be forcing, since incomparable element pairs are constrained into a ranking, thus losing information on the pattern and complexity of the input data structure. To maintain control over this “complexity-reducing” process, dominance scores are complemented with *incomparability scores* (as proposed in [54]) to assess the extent to which poset elements are incomparable with the others and thus the extent to which they are “rankable”. Since the higher $pdom_{ij}$, the more x_i and x_j are comparable and the less they are incomparable, it is natural to define the pairwise incomparability score between x_i and x_j as $pinc_{ij} = 2 \cdot \min(pdom_{ij}, pdom_{ji}) = 2 \cdot \min(D_{ij}, D_{ji})$ and arrange them into the $n \times n$ *incomparability matrix* N . By construction, N is symmetric, has null diagonal and its entries satisfy $0 \leq N_{ij} \leq 1$. Each poset element x_j is then assigned a global incomparability score inc_j by simply computing the sum of its pairwise incomparability scores in the corresponding column of N . The maximum incomparability score (i.e., $(n - 1)$) is attained by the elements of an antichain, while the minimum (i.e., 0) by the elements of a linear order, so that $inc_j^* = inc_j / (n - 1)$ is a normalized incomparability score.

Dominance and incomparability scores for the PP&P data. Table 4 reports the normalized dominance and incomparability scores computed on the PP&P poset, and Fig. 2 combines them into a scatter plot. As expected, the top (4444) and bottom (0000) profiles have dominance scores equal to 1 and 0, respectively, and both have null incomparability scores. As can be seen, the profiles are highly comparable (the only profile with incomparability score greater than 0.2 is 4331); the input data can thus be safely synthesized into the dominance scores that, once associated with the statistical units, produce the country rankings shown in the left column of Fig. 8 displayed at the end of the paper (each country is scored as the corresponding profile such that the final country ranking has several ties).

A technical remark. A quite natural alternative to the use of sums (or simple means) in the computation of dominance and incomparability scores is to perform the Singular Value Decomposition of matrices D and N and to take, as scores attached to profiles, the components

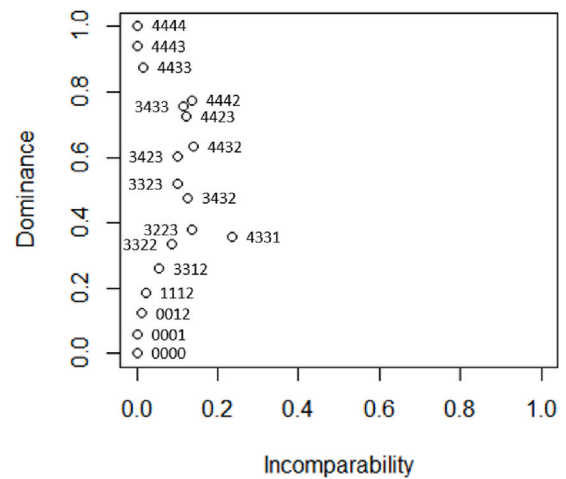


Fig. 2. Scatter plot of the normalized incomparability and dominance scores for the 18 unique profiles of the PP&P MIS.

Table 4

Unique PP&P profiles with corresponding normalized incomparability and dominance scores.

Prog.	1	2	3	4	5	6	7	8	9
Profile	4444	4443	4442	4433	4432	4423	4331	3433	3432
<i>inc</i> *	.000	.000	.136	.017	.141	.122	.235	.114	.125
<i>dom</i> *	1.000	.941	.773	.874	.632	.723	.356	.755	.476
Prog.	10	11	12	13	14	15	16	17	18
Profile	3423	3323	3322	3312	3223	1112	0012	0001	0000
<i>inc</i> *	.102	.101	.086	.055	.136	.021	.011	.000	.000
<i>dom</i> *	.603	.519	.337	.263	.379	.187	.123	.059	.000

of the first right-singular vector multiplied by the first singular value (see [55]) — that is, to define

$$dom_j = [\sigma_1 \mathbf{v}_1]_j = [\mathbf{u}_1^T D]_j \quad (5)$$

where σ_1 is the first singular value of D and $\mathbf{u}_1$ and $\mathbf{v}_1$ are the first columns of U and V in the decomposition $D = U \Sigma V^T$. The advantage of using the SVD is that the resulting score vector is “optimal”, being linked to the best one-rank approximation of D in the Frobenius norm. However, showing how to normalize this result proves to be more difficult. Since, in practice, using unweighted sums of the pairwise dominance degrees produces very similar results to using the SVD

approach, and since in the former case the properties of the scoring functions follow easily, we choose the first option here and for the remainder of the paper.

5. Scoring contexts and separation scores

The scoring procedure introduced in the previous section has been developed in the scope of MIS analysis because this is a typical setting in socioeconomic studies. A closer inspection, however, reveals that the indicator system plays no essential role here. Apart from providing the elements of the input poset, its only trace is in the choice of the component-wise ordering criterion that is by no means the only possible comparison rule (see Section 6). In fact, the scoring procedure proposed here is completely general and can be instantiated on any finite poset irrespective of its derivation from an MIS. To realize the specificity of posets deriving from MISes and the key role that these may have in the scoring process, consider again the Hasse diagram in Fig. 1. Profiles 4331 and 3432 are almost “equivalent” PP&P patterns, but the former dominates just two profiles and is comparable with seven, while the latter dominates six profiles and is comparable with 12. Such a “distortion”, which is ultimately reflected in the dominance and incomparability scores attached to them, arises since only the observed profiles have been considered; they disappear as soon as one realizes that 4331 and 3432 are embedded in a larger and mostly unobserved poset composed of all possible $5^4 = 625$ profiles constructed from the PP&P variables. This simple example shows that the “full” poset $\pi_{\mathbf{x}}^{full}$ derived from the input MIS provides the proper *scoring context* for the structural differences among profiles to be resolved or their similarities restored. Moreover, the existence of unobserved profiles “between” observed ones can be exploited to improve the discrimination power of the dominance scores by accounting for the information on the “quantity of profiles” between poset elements, as we do by introducing separation scores in the second part of this section.

5.1. Dominance and incomparability scores on the full poset

To account for the existence of unobserved profiles, the computation of dominance and incomparability scores is performed as follows:

1. build the component-wise poset $\pi_{\mathbf{x}}^{full}$ on the set of all profiles derived from the input variables;
2. compute the “embedded” pairwise dominance scores $f_{pdom_{ij}}$ between all element pairs in the full poset and, from these, build the dominance and incomparability matrices D^{full} and N^{full} ;
3. from D^{full} and N^{full} , build the observed matrices D^{obs} and N^{obs} by extracting the submatrices corresponding to the observed profiles;
4. compute the embedded dominance and incomparability scores f_{dom_j} and f_{inc_j} for all of the observed profiles, by summing the columns of D^{obs} and N^{obs} ;
5. compute the normalized embedded scoring functions as $f_{dom_j}^* = f_{dom_j}/(n-1)$ and $f_{inc_j}^* = f_{inc_j}/(n-1)$.

In practice, the underlying logic is the same as in the decontextualized case, but the pairwise scores to be synthesized are computed within the full poset.

Incomparability and dominance scores on the full PP&P poset. The left panel of Fig. 3 depicts the scatter plot of the incomparability and dominance scores for the PP&P dataset computed in the full poset and reported in Table 5. The effect of the “embedding” is quite neat: the final profile rankings change mostly in the middle of the distribution (see the second column of Fig. 8), and the range of the embedded incomparability scores is narrower as a consequence of the presence of many unobserved profiles that better resolve the dominance between pairs of incomparable profiles (notice, however, that some profiles – e.g., 4433 – increase their incomparability scores).

Table 5

Unique PP&P profiles with corresponding normalized embedded f_{inc}^* , f_{dom}^* and f_{sep}^* scores (the incomparability scores of profiles 1112 and 0012 are very small and thus approximated to 0.000).

Prog.	1	2	3	4	5	6	7	8	9
Profile	4444	4443	4442	4433	4432	4423	4331	3433	3432
f_{inc}^*	.000	.000	.076	.047	.124	.136	.145	.111	.107
f_{dom}^*	1.000	.941	.844	.847	.703	.710	.453	.696	.548
f_{sep}^*	1.000	.981	.942	.937	.873	.873	.664	.862	.772

Prog.	10	11	12	13	14	15	16	17	18
Profile	3423	3323	3322	3312	3223	1112	0012	0001	0000
f_{inc}^*	.102	.068	.072	.021	.104	.000	.000	.000	.000
f_{dom}^*	.554	.446	.330	.246	.329	.177	.118	.059	.000
f_{sep}^*	.772	.645	.533	.421	.533	.096	.020	.002	.000

5.2. Separation scores

When measuring the effective quantity of poset elements below a given profile, dominance scores provide a natural criterion for reliable rankings (and considering the *b/s* function as an approximation to mutual ranking probabilities, they essentially correspond to the so-called *average ranks*; see [52]). However, as can be seen in the scatter plots displayed herein, they also tend to distribute profiles too uniformly along the vertical axis. This is because pairwise dominance scores capture no information on, informally speaking, the “distance” between elements in the full poset, thus lessening the polarization and discrimination power of the final synthetic scores. To overcome this issue, we search for new pairwise scores $f_{psep_{ij}}$ by measuring the number of profiles “below $\mathbf{x}_j$ and above $\mathbf{x}_i$ ”. In a poset, however, this is not entirely trivial, since for two incomparable elements $\mathbf{x}_i$ and $\mathbf{x}_j$, no element $\mathbf{x}_k$ such that $\mathbf{x}_i < \mathbf{x}_k < \mathbf{x}_j$ exists. However, since $f_{pdom_{ik}}$ and $f_{pdom_{kj}}$ express the strengths of the statements $\mathbf{x}_i < \mathbf{x}_k$ and $\mathbf{x}_k < \mathbf{x}_j$ in the full poset, their product $f_{pdom_{ik}} f_{pdom_{kj}}$ can be seen as the strength of the statement $\mathbf{x}_i < \mathbf{x}_k < \mathbf{x}_j$. Consequently, the *pairwise separation score* is calculated as

$$f_{psep_{ij}} = f_{pdom_{ij}} + \sum_{k=1}^{n^{full}} f_{pdom_{ik}} f_{pdom_{kj}} \quad (6)$$

(where n^{full} is the cardinality of the full poset) and can be understood as the “effective” quantity of elements “below $\mathbf{x}_j$ and above $\mathbf{x}_i$ ” (the term $f_{pdom_{ij}}$ has been added since in $f_{psep_{ij}}$ we want to take into account $\mathbf{x}_i$ itself). Notice that $f_{psep_{ij}}$ is not symmetric in i and j . By computing pairwise separation scores for each pair of poset elements and arranging them into a matrix, we obtain the *separation matrix* S^{full} , which is computed from the dominance matrix D^{full} as $S = (I + D^{full})D^{full}$ and the observed separation matrix S^{obs} obtained from S^{full} by considering the row and column indices corresponding to the observed profiles. By summing the pairwise separation scores of $\mathbf{x}_j$ relative to the observed profiles, we obtain the separation score

$$f_{sep_j} = [1_n^T S^{obs}]_j, \quad (7)$$

which can be normalized as

$$f_{sep_j}^* = \frac{f_{sep_j}}{f_{sep_T}}, \quad (8)$$

where f_{sep_T} is given by

$$f_{sep_T} = \sum_{i=1}^n f_{psep_{iT}}, \quad (9)$$

assuming $\mathbf{x}_1, \dots, \mathbf{x}_n$ to be the observed profiles.

Separation scores for the PP&P data. The right panel of Fig. 3 shows that the discrimination effect is enhanced by the separation scores when compared to the dominance scores (for numerical figures, see Table 5). Now, profiles 1112, 0012, 0001 and 0000 are neatly separated from the others, which more clearly reflects their intrinsically lower PP&P

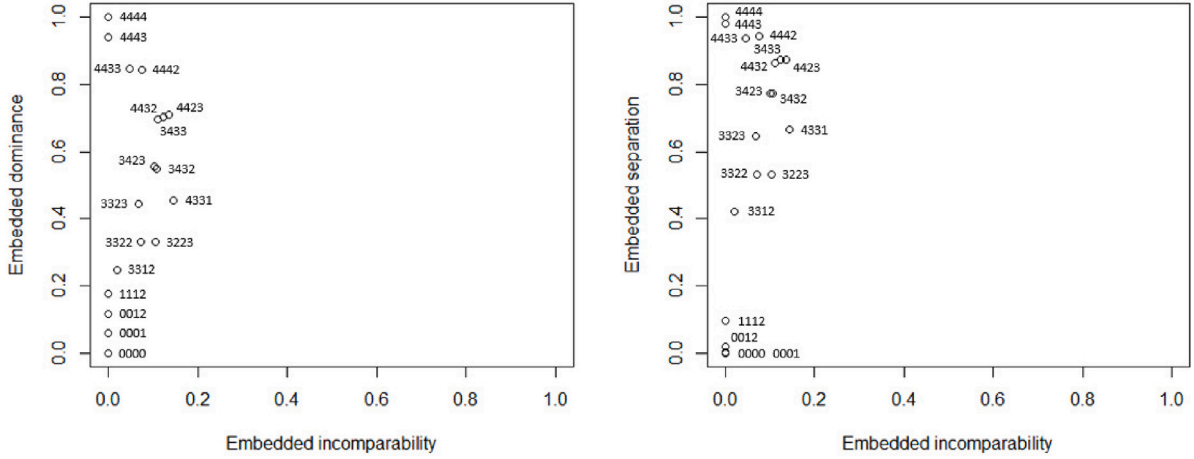


Fig. 3. Scatter plots of the incomparability, dominance and separation scores $finc$, $fdom$ and $fsep$, for the unique PP&P profiles embedded in the full PP&P poset.

level. As is made clear by comparing the second and third columns of Fig. 8, the rankings delivered by the dominance and separation scores are almost identical, but the “metric” information of the latter makes them definitively more informative when the distribution of the scores along the vertical axis is considered.

6. Accounting for variable importance

A typical and often critical issue in indicator construction is accounting for the different importances of the input variables in the synthesis process. Usually, this is done by attaching numerical weights to the variables based on exogenous experts’ judgments, as often in the composite indicators approach, or on the covariance structure of the data, as in Principal Component Analysis. When dealing with ordinal MISes, this is obviously unfeasible as mixing ordinal scores with numbers is not meaningful, and thus other ways of incorporating variable importance must be devised. Furthermore, another and more fundamental issue makes the use of numerical weights problematic. Being linearly ordered, numbers may well describe cases where one variable is less, equally or more relevant than another, but fail to reproduce situations where two variables cannot be ranked in terms of their relative importance. Indeed, a policymaker could legitimately deem the PP&P variable B1 to be more important than the others and variable B3 to be more important than B4 without expressing any priority between B2 and B3 or B2 and B4. Thus, variable importance is more naturally and flexibly represented as a poset rather than as a (possibly tied) ranking or through metric scales. If we accept this point of view, the issue of how the information comprised in the partial order relation set on the variables is transferred into the computation of the scores arises. The key point here is that in the posetic approach, this depends upon the structure of the partial order relation put on the MIS profiles and not upon the unrelated set of their internal configurations. Variable importance is thus introduced into the computations by properly modifying the component-wise input poset π_x associated with the MIS according to the partial order relation put on the variables. For example, assuming B3 is more relevant than B4, the incomparability 4432||4423 in π_x is naturally turned into the dominance 4423<4432, thus reshaping the original poset into a new partially ordered structure. After adding all the dominances implied by the variable poset to the profile poset, the score computation may proceed as before with the new poset being used as the input. We now formalize this process and introduce the necessary order-theoretic concepts.

Let $\pi_1 = (X, \leq_1)$ and $\pi_2 = (X, \leq_2)$ be two posets on the same set; π_2 is a (strict) *extension* of π_1 , denoted as $\pi_1 \subset \pi_2$ if and only if $x_i \leq_1 x_j \Rightarrow x_i \leq_2 x_j$ and $x_s, x_t \in X$ exist such that $x_s \parallel_1 x_t$ and $x_s <_2 x_t$ (i.e., if π_2 comprises the same set of dominances of π_1 and

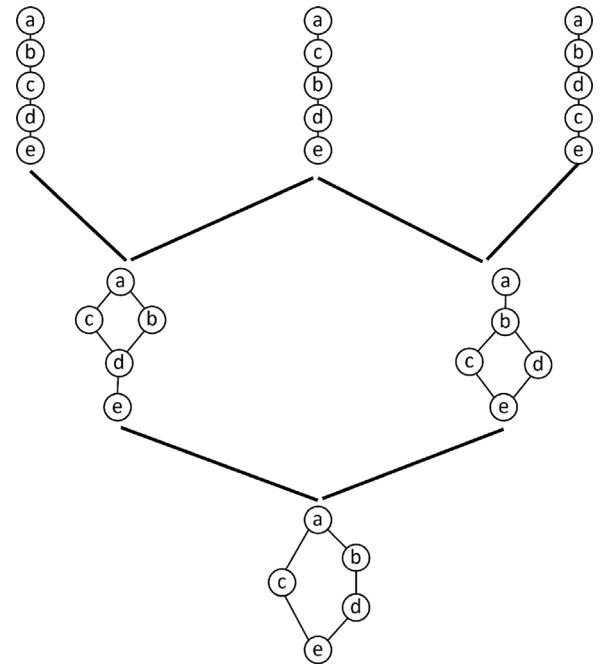


Fig. 4. Hasse diagram of the extension poset of a poset with five elements.

at least one more; notice that if π_2 is a linear order, we obtain a linear extension; see paragraph 4.1). The inclusion of binary relation $\subseteq$, where $\pi_1 \subseteq \pi_2 \Leftrightarrow \pi_1 \subset \pi_2$ or $\pi_1 = \pi_2$, is easily seen to be a partial order; under it, the set $Ext(\pi)$ of all extensions of a poset π is a itself a poset, with π as the bottom and with the linear extensions of π as maximal elements (see Fig. 4, for a simple example). Notice that simply adding dominances to π generally does not yield a new poset (and thus an extension) because all dominances implied by the added comparabilities, through transitivity, must be added as well (i.e., taking the so-called *transitive closure*; see [56]).

Let Θ be the set of all posets over the variables $V_1, \dots, V_k$ of the input MIS and let $\theta \in \Theta$ be one such poset. Our goal is to define an *extension map* $\gamma : \Theta \rightarrow Ext(\pi_x)$ that picks up an extension $\pi_\theta = \gamma(\pi_x|\theta)$ of π_x , thus exploiting the information comprised in θ . We restrict the class of the extension maps by imposing the following requirements:

1. θ antichain $\Rightarrow \pi_\theta = \pi_x$;
2. $\theta_1 \subseteq \theta_2 \Rightarrow \pi_{\theta_1} \subseteq \pi_{\theta_2}$;

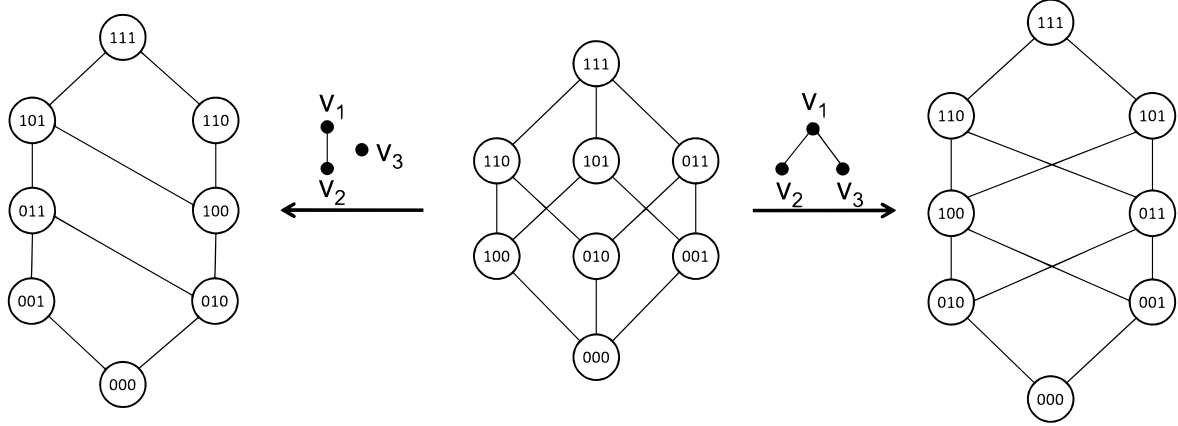


Fig. 5. Two examples of extensions of poset 2^3 (i.e., the “cube”), which is built on 3 binary variables, corresponding to the variable posets depicted above the arrows.

3. let θ comprise just one nontrivial dominance $V_i \triangleleft_{\theta} V_j$ and let $\mathbf{x}_s, \mathbf{x}_t \in X$ be such that $\mathbf{x}_{ti} < \mathbf{x}_{si}$, $\mathbf{x}_{sj} < \mathbf{x}_{tj}$ and $\mathbf{x}_{sh} \leq \mathbf{x}_{th}$ for $h \neq i, j$. Then, $\mathbf{x}_s \triangleleft \mathbf{x}_t$ in π_{θ} .

Property 1 simply states that if no relative importance among variables is set, then γ is the identity map — that is, the input poset $\pi_{\times}$ is unaffected by the extension process. Property 2 requires γ to be (weakly) order-preserving; indeed, if θ_2 adds some variable dominances to θ_1 , then π_{θ_2} must comprise at least the same dominances of π_{θ_1} . Property 3 defines the action of the extension map on the most “elementary” nontrivial variable posets: if the only dominance in θ involves V_i and V_j , then profiles having conflicting scores only on those variables are ordered accordingly in π_{θ} . An extension map satisfying these properties is here called *admissible*; the class of admissible maps is denoted by $\Gamma(\Theta)$.

Is there any “optimal” map in Γ ? The natural choice is to search for the map (if it exists) that adds to $\pi_{\times}$ the minimum subset of dominances given θ or, differently stated, that provides the most “conservative” extension, where only dominances “strictly” implied by θ are inserted. The Appendix shows that $\Gamma(\Theta)$ can be naturally structured as a poset having a bottom element $\gamma_{\perp}$, that thus satisfies the required minimality property. The action of $\gamma_{\perp}$ is very simple: for every variable dominance $V_i \triangleleft_{\theta} V_j$ in θ : (i) add to $\pi_{\times}$ all and only the profile dominances $\mathbf{x}_s \triangleleft \mathbf{x}_t$ where $\mathbf{x}_s$ and $\mathbf{x}_t$ conflict only on V_i and V_j , with $\mathbf{x}_{ti} < \mathbf{x}_{si}$ and $\mathbf{x}_{sj} < \mathbf{x}_{tj}$, and (ii) take the transitive closure of the resulting dominance set. Fig. 5 provides some examples of the action of $\gamma_{\perp}$ on a small poset built on three binary variables.

Scores for the extended PP&P poset. To show the extension map $\gamma_{\perp}$ in action, we first apply it to the original observed PP&P poset $\pi_{\times}$ depicted in Fig. 1, with variable poset $\theta = B4 \triangleleft_{\theta} \{B1, B2, B3\}; B1 \parallel B2 \parallel B3$ (i.e., variable B4 is less important than B1, B2 and B3, all of which are equally important) to obtain the extended Hasse diagram depicted in Fig. 6. As can be seen, the resulting poset π_{θ} is now more similar to a linear order comprising 146 comparabilities and only 7 incomparabilities as an effect of the added dominances. We then compute the normalized incomparability, dominance and separation scores (*extfinc**, *extfdom** and *extfsep**, respectively) and embed the observed profiles in the full extended poset π_{θ}^{full} , whose Hasse diagram is too large to be depicted (since the extension process is computationally heavy, we uniformly sampled 100 profiles from $\pi_{\times}^{full}$ in addition to the 18 observed samples, thus extending the resulting poset by 118 component-wise ordered elements). The (approximated) results are plotted in Fig. 7 and numerically reported in Table 6. In addition to the higher discrimination power of the separation scores, it can be clearly seen that the incomparability scores are now very low, as expected, thus making the final ranking even more reliable. The right column of Fig. 8 provides the country rankings based on the separation scores, which has some differences compared to the non-extended scores and shows that the

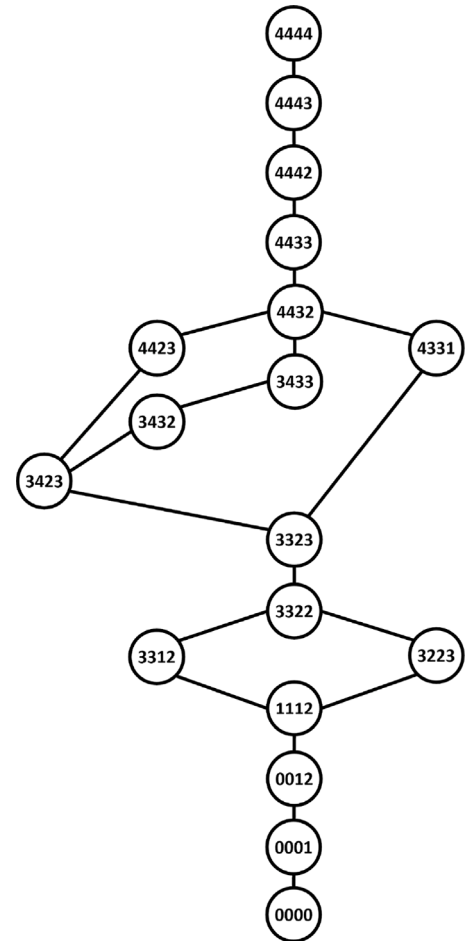


Fig. 6. Hasse diagram of the unique PP&P profiles poset, extended according to the variable poset $\theta = B4 \triangleleft_{\theta} \{B1, B2, B3\}; B1 \parallel B2 \parallel B3$ given in the main text.

introduction of variable importance may well affect the final position of the units.

7. Scores on weighted posets

In practical applications, profiles are often naturally assigned absolute or relative frequencies and modifying the scoring procedure to account for them is straightforward. Let $\mathbf{w} = (w_1, \dots, w_n)$ be the frequency

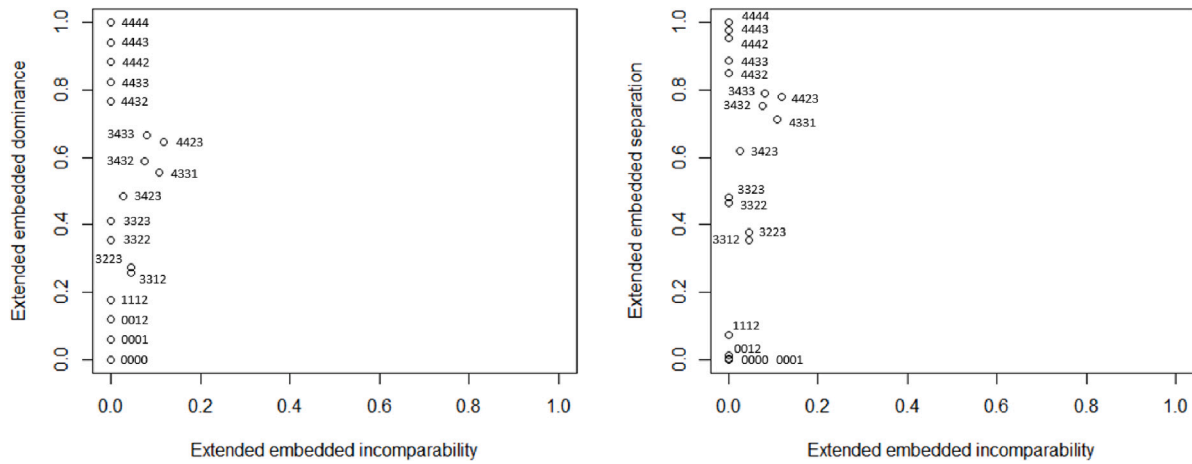


Fig. 7. Scatter plot of the normalized incomparability, dominance and separation scores for the (sampled) extended full PP&P poset.

Table 6

Unique PP&P profiles with corresponding normalized extended scores computed in the full poset.

Prog.	1	2	3	4	5	6	7	8	9
Profile	4444	4443	4442	4433	4432	4423	4331	3433	3432
<i>extfinc*</i>	.000	.000	.000	.000	.000	.102	.121	.069	.063
<i>extfdom*</i>	1.000	.991	.976	.952	.920	.816	.780	.849	.750
<i>extfsep*</i>	1.000	.976	.954	.886	.851	.780	.712	.790	.754
Prog.	10	11	12	13	14	15	16	17	18
Profile	3423	3323	3322	3312	3223	1112	0012	0001	0000
<i>extfinc*</i>	.013	.000	.000	.588	.588	.000	.000	.000	.000
<i>extfdom*</i>	.665	.590	.514	.393	.393	.267	.179	.090	.000
<i>extfsep*</i>	.620	.482	.463	.354	.376	.072	.013	.001	.000

vector and $W = \text{diag}(w_1, \dots, w_n)$ be the diagonal weighting matrix. In the more general case where the full poset is considered (the decontextualized case for dominance and incomparability scores is recovered as a special instance), weighted dominance, incomparability and separability matrices are obtained as WD^{obs} , WN^{obs} and WS^{obs} , along with the corresponding weighted scoring functions $wfdom$, $wfinc$ and $wfsep$ that are computed as in the unweighted case, using such matrices as input. The weighted incomparability and dominance scores can be normalized by dividing them by $(1 - 1/n) \sum_{i=1}^n w_i$ and the separation scores by $[(1 - 1/n) \sum_{i=1}^n w_i] \cdot fpsep_{\perp}$. The latter takes as a reference the case where $w_{\top} = 1$ and $w_{\perp} = n - 1$ (or $1/n$ and $1 - 1/n$, in the relative case) — that is, the most separated distribution. This is, however, an extreme and often unrealistic weight configuration that tends to squeeze the final scores, and more reasonable benchmarks can be chosen on a case-by-case basis. When variable importance is introduced into the computations, the weighted version of the scores is recovered by substituting D^{obs} , N^{obs} and S^{obs} above with the corresponding matrices D_{θ}^{obs} , N_{θ}^{obs} and S_{θ}^{obs} , which are computed on the extended poset π_{θ} .

8. Discussion and comments

In this section, we provide a general discussion on the proposed scoring/ranking procedure, highlight its originality and identify its possible limitations. We start by summarizing the novel contributions of this paper as follows:

1. Separation scores are introduced as the key quantities for scoring and ranking units. This represents an improvement to the previous literature, which mainly relies on average ranks and dominance scores [6,38,39,55]. These scores do not account for the “distance” between units in the input poset and thus

only exploit the information comprised in it in a limited way. Separation scores, on the other hand, reflect the “amount” of space between pairs of elements and thus deliver much more informative outputs, as shown in the Political Pluralism and Participation example.

2. The fundamental distinction between “generic posets” and posets that are derived from multi-indicator systems is often absent in previous literature, but emphasized and exploited here, in the design of the scoring procedure. This is a key point because the input MIS represents the context in which to set the scoring process; as such, it provides essential additional information that procedures designed for generic posets cannot exploit, as discussed in the main text. Most applied literature employing posetic algorithms, however, overlooks this issue (see, e.g., nevertheless interesting papers such as [43,57–59]).
3. A new way to account for variable importance that is more flexible and effective compared to existing techniques is designed and shown in action in this paper. This is relevant in making the scoring procedure practically useful in evaluation processes and multi-criteria decision-making, as both may depend upon external judgments, “value systems”, or “preferences”... that require the ability to alter the basic comparison criterion. The idea of setting a partial order relation on the variables and connecting it to the extension of the component-wise poset has already been put forward in [45]; however, the proposed procedure there is too rigid and imposes too many additional dominances on the input poset. This does not occur with the weak extension procedure designed here, which allows for a softer and more robust way of introducing partial priorities among variables.

As for practical applications and potential policy implications, the procedures designed in this paper are relevant in view of the evaluation and monitoring of SDGs and in the so-called “beyond GDP” approach, where monetary measures of societal wealth are substituted with more nuanced and softer measurement criteria. In particular, separation scores and the weak extension algorithm render the entire procedure flexible and effective and offer a different way of satisfying policymakers’ needs for synthetic information. On the one hand, the discrimination power of separation scores, which is confirmed by the final plots of the Political Pluralism and Participation example, may resolve differences between units that dominance scores (and even more aggregated indicators) may not. The extension algorithm, on the other hand, makes it easier to address the critical problem of accounting for variable importance not only from a technical point of view (i.e., demonstrating how to do this in an ordinal context), but, perhaps more importantly, by substituting numerical weights (the usual way to introduce variable priorities into socioeconomic indicators) with

softer ordinal judgments. While the choice of the weighting system is typically a controversial, sometimes artificial, and problematic step in indicator construction, the choice of a partial order to be put on the set of variables is likely to be a more reliable and easy-to-agree-upon task.

The proposed algorithms also have some limitations. The combinatoric nature of the computations involved in the scoring and ranking processes are quite heavy and may limit the complexity of the posets that can be efficiently handled. In particular, the extension procedure that accounts for variable importance is very demanding and may require sampling approximations, although software implementations in efficient programming languages (e.g., C++) may lessen this issue. A second and more theoretical limitation is that, unlike dimensionality reduction for multidimensional numerical data, it remains unclear how to extract further “components” beyond the synthesis achieved using dominance or separation scores. In other words, it is unclear how to search for multidimensional approximations of the input poset. Rather than being an algorithmic problem, this is a mathematical issue that requires developing suitable order-theoretic results to be later turned into computational procedures. A third limitation pertains to the interpretation of the final scores; this should be based on their link to the input variables, but no tool for inspecting this point has been developed yet.

9. Conclusions

In this paper, we have designed a coherent and comprehensive framework for building synthetic sustainability indicators and rankings that preserve and reinterpret the classical logic of indicator construction in a multidimensional ordinal setting. The input ordinal data are turned into a partially ordered set and the computational grammar of order theory is employed to extract dominance, incomparability and separation scores, thus overcoming the limitations of aggregative-compensative procedures and the need for scaling algorithms. It has been also shown, in a purely ordinal way, how information on variable importance can be easily introduced into the computations, thus making the scoring process flexible and adaptable to different aims and contexts. The procedures have been shown in action on a real dataset pertaining to Political Pluralism and Participation in North and South American countries.

We also suggest some research and development opportunities, as outlined below:

1. The scope of the scoring and ranking toolbox should be widened to account for targeting and tracking. This requires introducing distinct elements playing the role of targets or thresholds into the input poset and computing the ranking scores making reference to them. This procedure could then be used to monitor the achievement of the sustainability goals over time.
2. A general theory of dominance functions and dominance matrices is needed to more firmly establish the basis of the posetic approach and to potentially address the computational issues that unavoidably arise when dealing with combinatorial structures. Similarly, working out alternative extension criteria to account for variable importance and putting them in a coherent and comprehensive framework would be a valuable step forward, in developing applications to multi-criteria decision-making.
3. As mentioned in the previous section, there are currently no algorithms that extend the scoring process to produce bi-dimensional (or, in general, higher dimensional) reduced representations of the input poset in the spirit of what is done by extracting more components in principal component analysis. The main problem here is that, at least to our knowledge, there is no easy way to introduce the concept of “orthogonality” (informally speaking) in order theory and thus to decompose ordered structures into orthogonal components. This is surely a relevant research topic, albeit one of a mathematical rather than a statistical nature.

In addition to providing algorithms for indicator construction, this paper also aimed at fostering an “ordinal” and “soft modeling” approach to evaluation and ranking. While rankings and synthetic indicators are gaining relevance in decision-making in many contexts, the issues of reliability and interpretability often arise. This is in contrast to the tendency of mimicking and pursuing the precision of physical measurement, which is often observed in the socioeconomic sciences. However, socioeconomic traits are often nuanced, fuzzy in nature and difficult to trace back to formal definitions. As a result, forcing them into the classical framework of statistical measurement can be at odds with the nature of the concepts being assessed and thus a soft and ordinal approach may be preferable and prove to be more consistent and reliable than “hard” measurement procedures (see Sen’s deep discussion of this point of view, in the context of poverty and inequality assessment [9]). In this respect, it is worth noting that order theory may play a role in multidimensional data analysis beyond the ordinal case. Partial order relations appear wherever incomparabilities are of interest and variable aggregation is conceptually problematic, as in the assessment of sustainability or human development [43], in social studies [60], in environmental and ecological analysis [42], and in all cases where multifaceted and nuanced phenomena are to be statistically described and measured.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix. Minimality of $\gamma_{\perp}(\theta)$

In this Appendix, we prove the optimality property of the extension map $\gamma_{\perp}(\theta)$ stated in the main text.

Let Θ be the collection of all posets over the set of variables $\{V_1, \dots, V_k\}$ and let $\gamma_1, \gamma_2 \in \Gamma(\Theta)$ be two admissible extension maps. The binary relation $\trianglelefteq_{\Gamma(\Theta)}$ defined by $\gamma_1 \trianglelefteq_{\Gamma(\Theta)} \gamma_2$ if and only if $\gamma_1(\pi_{\times}|\theta) \subseteq \gamma_2(\pi_{\times}|\theta)$ for all $\theta \in \Theta$ is a partial order over $\Gamma(\Theta)$. For $\theta \in \Theta$, let Δ_{θ} be the set of dominances between profiles in $\pi_{\times}$ directly implied by θ , as defined in Section 6, and let $\gamma_{\perp}$ be the extension map defined by

$$\gamma_{\perp}(\pi_{\times}|\theta) = \overline{\{\pi_{\times} \cup \Delta_{\theta}\}} \quad (10)$$

where $\overline{}$ denotes the transitive closure operation.

To show that $\gamma_{\perp}$ is the minimum of $\Gamma(\Theta)$, let $\gamma \in \Gamma(\Theta)$ be a generic admissible extension map, let $Q_{\theta} = \{(i, j) : V_i \triangleleft_{\theta} V_j\}$ be the set of index pairs corresponding to the strict dominances of θ and let θ_{ij} be the variable poset whose only strict dominance is $V_i \triangleleft_{\theta_{ij}} V_j$ (with $(i, j) \in Q_{\theta}$). By Properties 2 and 3 of the extension maps (see Section 6), it follows that $\gamma(\pi_{\times}|\theta_{ij}) \subseteq \gamma(\pi_{\times}|\theta)$ for any pair $(i, j) \in Q_{\theta}$ so that $\gamma(\pi_{\times}|\theta)$ comprises all of the dominances in $\pi_{\times}$ and those in Δ_{θ} (i.e., $\{\pi_{\times} \cup \Delta_{\theta}\} \subseteq \gamma(\pi_{\times}|\theta)$). By taking the transitive closure of both sides, we obtain:

$$\gamma_{\perp}(\pi_{\times}|\theta) = \overline{\{\pi_{\times} \cup \Delta_{\theta}\}} \subseteq \overline{\gamma(\pi_{\times}|\theta)} = \gamma(\pi_{\times}|\theta) \quad (11)$$

because $\gamma(\pi_{\times}|\theta)$ satisfies the transitivity property, being a poset. Thus $\gamma_{\perp} \trianglelefteq_{\Gamma(\Theta)} \gamma$ and since γ is arbitrary, $\gamma_{\perp}$ is the minimum of $\Gamma(\Theta)$. $\square$

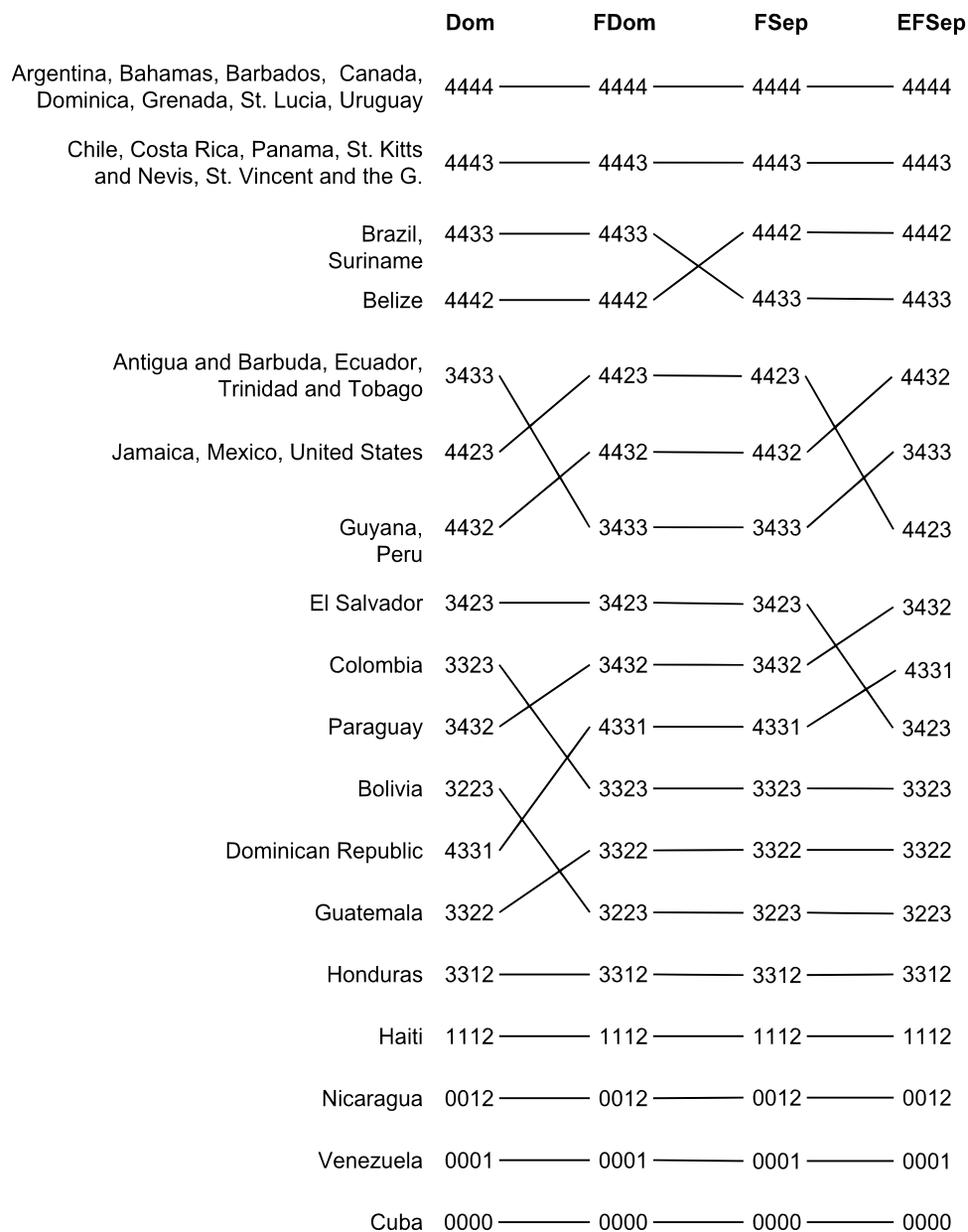


Fig. 8. Comparison of the country rankings according to the normalized dominance scores computed on the observed poset π_x (Dom), on the full component-wise poset π_x^{full} (FDom), according to separation scores computed on π_x^{full} (FSep) and on the extended full poset π_θ^{full} (EFSep).

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Marco Fattore is Associate Professor of Economic Statistics, at the Department of Statistics and Quantitative Methods of University of Milano-Bicocca. He holds a degree in Theoretical Physics and a Ph.D in Statistics. His main research interest lies in the interplay among order theory, multidimensional analysis of ordinal data systems and computational Statistics, for the construction of synthetic indicators and the design of algorithms for the measurement and the assessment of socio-economic phenomena, like deprivation, sustainability, well-being and the like.

Leonardo Salvatore Alaimo is Research Fellow of Social Statistics (SECS-S/05) at the Department of Social Sciences and Economics of University of Rome La Sapienza. He holds a Ph.D. in Social Statistics at University of Rome La Sapienza. He was a researcher at the Italian National Institute of Statistics – Istat and expert on Statistics of the Italian Prime Minister Office – Italian Government (Conte's cabinet). His research interests are mainly the measurement of complex phenomena, the synthesis of systems of statistical indicators and the multivariate statistical methods.